

It yields a criterion that is very easy to check “by inspection” for the interesting case $n - d = 4$; see below.

Corollary 6.20 (Characterization of Gale diagrams of polytopes).

A configuration $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ points in $(\mathbb{R}^{n-d-2})^*$, each of them declared to be either “positive” or “negative,” that affinely spans $(\mathbb{R}^{n-d-2})^*$, is the Gale diagram of a d -polytope with n vertices if and only if the following condition is satisfied: for every oriented hyperplane H in $(\mathbb{R}^{n-d-2})^*$ spanned by points of A , the number of positive A -points on the positive side of H , plus the number of negative A -points on the negative side of H , is at least 2.

In our descriptions we have disregarded the case of “special” points: they are just the cone points, so adding k special points to the diagram G corresponds to taking the k -fold pyramid over the polytope represented by G .

6.5 Polytopes with Few Vertices

Any d -polytope with $d + 1$ vertices is a d -simplex: this we know and have seen before. In this case the Gale diagram is in 0-dimensional space, so all vectors are $\mathbf{0}$ -vectors trivially.

Next consider the case of d -polytopes with $d + 2$ vertices. The result is that there are $\lfloor d^2/4 \rfloor$ combinatorial types of d -polytopes with $d + 2$ vertices. Of those, $\lfloor d/2 \rfloor$ represent simplicial polytopes, and the others are (multiple) pyramids over simplicial polytopes of this type. The case of $d + 2$ vertices is classic and can be found in Schoute [480] and Sommerville [506]; see also Grünbaum [252, Sect. 6.1] or Ewald [201, Sect. 2.6].

The affine Gale diagrams representing d -polytopes with $d + 2$ vertices are 0-dimensional and may be represented by a “cloud” of positive points (black), negative points (white), and special points (grey). The condition of Corollary 6.20 requires that there are at least 2 black and at least 2 white points. Furthermore, interchanging black and white points does not change the combinatorial type of the polytope.

Thus we get the following complete enumeration for $d = 3$, $n = 5$:



represents the bipyramid over a triangle (this is a simplicial polytope with 6 facets), and



yields the square pyramid (a nonsimplicial polytope with 5 facets).

Similarly, you should analyze and classify the 4-polytopes with 6 vertices in Exercises 6.8.

For polytopes with $d + 3$ vertices, the complete classification becomes difficult, but not out of reach. Their affine Gale diagrams are configurations of signed points on a line. The main problem is to decide which different diagrams represent combinatorially equivalent polytopes. First results were due to Gale, while Perles developed the (Gale diagram) techniques necessary to analyze polytopes with $d + 3$ vertices. The special case of simplicial polytopes was done in Grünbaum [252, Sect. 6.2]; the work was completed by Mani [376] and Kleinschmidt [333]. Explicit formulas for the number of d -polytopes with $d + 3$ vertices were obtained by Perles [252] for the simplicial case and by Fusy [217] for the general case (correcting an error in an earlier solution by Lloyd [370]). Note that the octahedra that we considered before have $d + 3$ vertices, for $d = 3$.

The polytopes with $d + 3$ vertices still do not have any unusual properties.

Finally, we arrive at the case of polytopes with $d + 4$ vertices: here is the *threshold for counterexamples*, as Sturmfels [532] calls it. They can be analyzed in terms of planar point configurations — which can be arbitrarily complicated. In particular, for high enough d there are

- d -polytopes with $d + 4$ vertices that do not have a realization with rational coordinates,
- d -polytopes with $d + 4$ vertices for which the shape of a facet cannot be prescribed, and
- d -polytopes with $d + 4$ vertices that have disconnected realization spaces.

We will now describe Gale diagram approaches to these three phenomena.

(a) A Nonrational 8-Polytope

Using Gale diagrams, Perles has shown that there are nonrational polytopes, that is, polytopes for which there are no combinatorially equivalent polytopes with rational coordinates.

Example 6.21 (Perles). [252, p. 95] [96, Fig. 8.4.1]

There is a nonrational 8-polytope with 12 vertices.

To see this, one verifies that the configuration G in the figure on the next page has three properties:

1. G cannot be realized with rational coordinates “as a matroid”: there is no rational planar configuration of 12 points such that the same sets of points as in G coincide, respectively are collinear. (Essentially, there is a golden ratio involved in the construction of a regular pentagon, so it can only be realized with coordinates in a field containing $\sqrt{5}$.)